If you have an argument where $P_1, P_2, \ldots\ldots, P_r$ are the premises and C is the conclusion, to get a proof using resolution principle, Put $P_1, \ldots\ldots, P_r$ in clause form and add to it $\neg C$ in clause form. From this sequence, if $\square$ can be derived, the argument is valid.

For example, consider modus ponens

$$\begin{array}{l} P \\ P \to Q \\ \hline \therefore Q \end{array}$$

In the clause form $C_1 = P \quad C_2 = \neg P \vee Q$

Take as C_3, the negation of the conclusion, i.e. $\neg Q$, $C_3 = \neg Q$

From C_1 and C_2 by resolution you can get Q and from this and C_3 by resolution is arrived at.

$C_1 \quad P$

$C_2 \quad \neg P \vee Q$

$C_3 \quad \neg Q$

from C_1 and C_2 by resolution $C_4 \quad Q$

from C_3 and C_4 by resolution $C_5 \quad \square$

Example 11: We show that the following argument is correct.

If today is Tuesday, I have a test of Mathematics or Economics. If my Economics Professor is sick, I will not have a test of Economics. Today is Tuesday and my Economics Professor is sick. Therefore I have a test of Mathematics.

T: Today is Tuesday M: I have a test of Mathematics

E: I have a test of Economics P: My Economics Professor is sick.